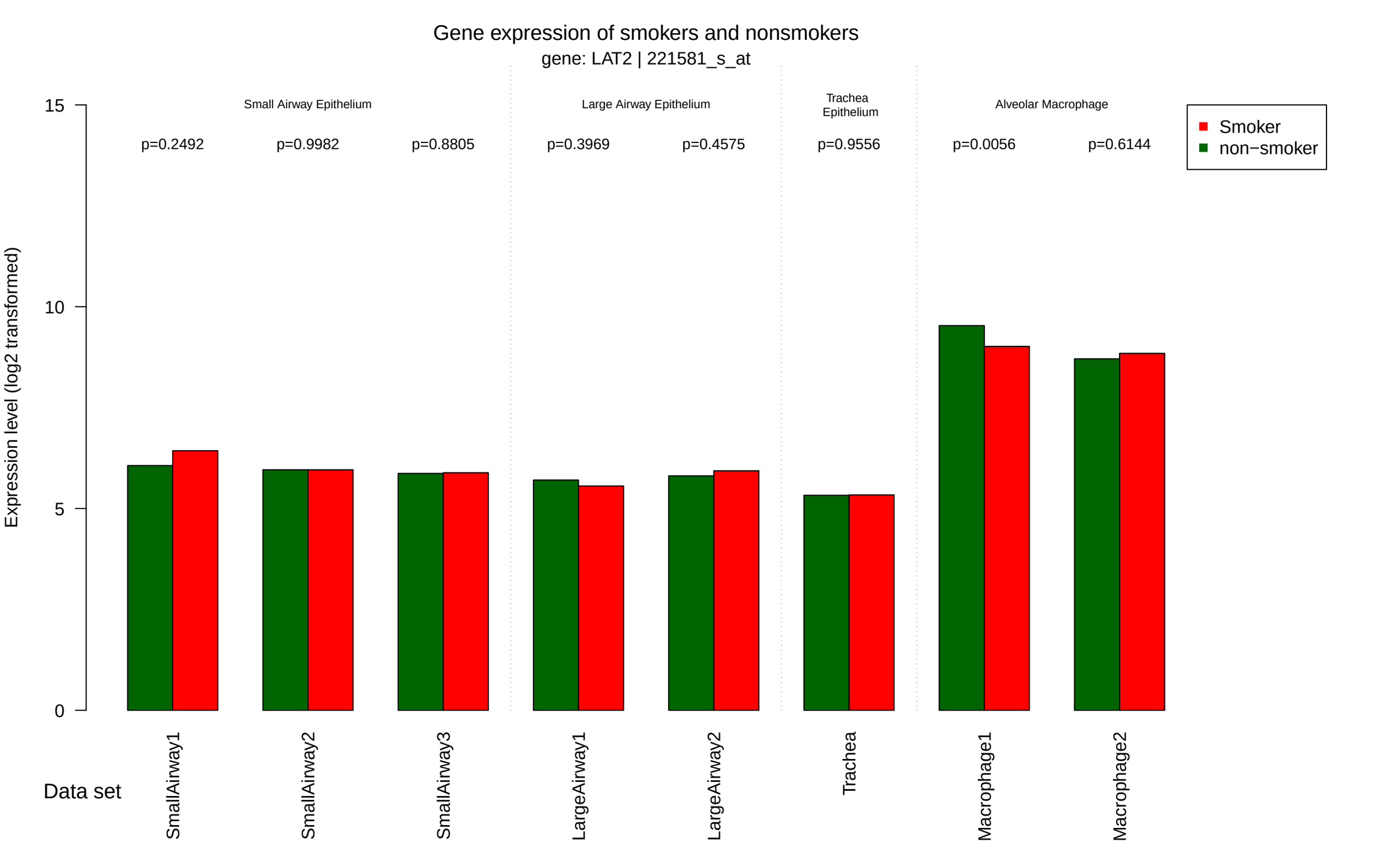
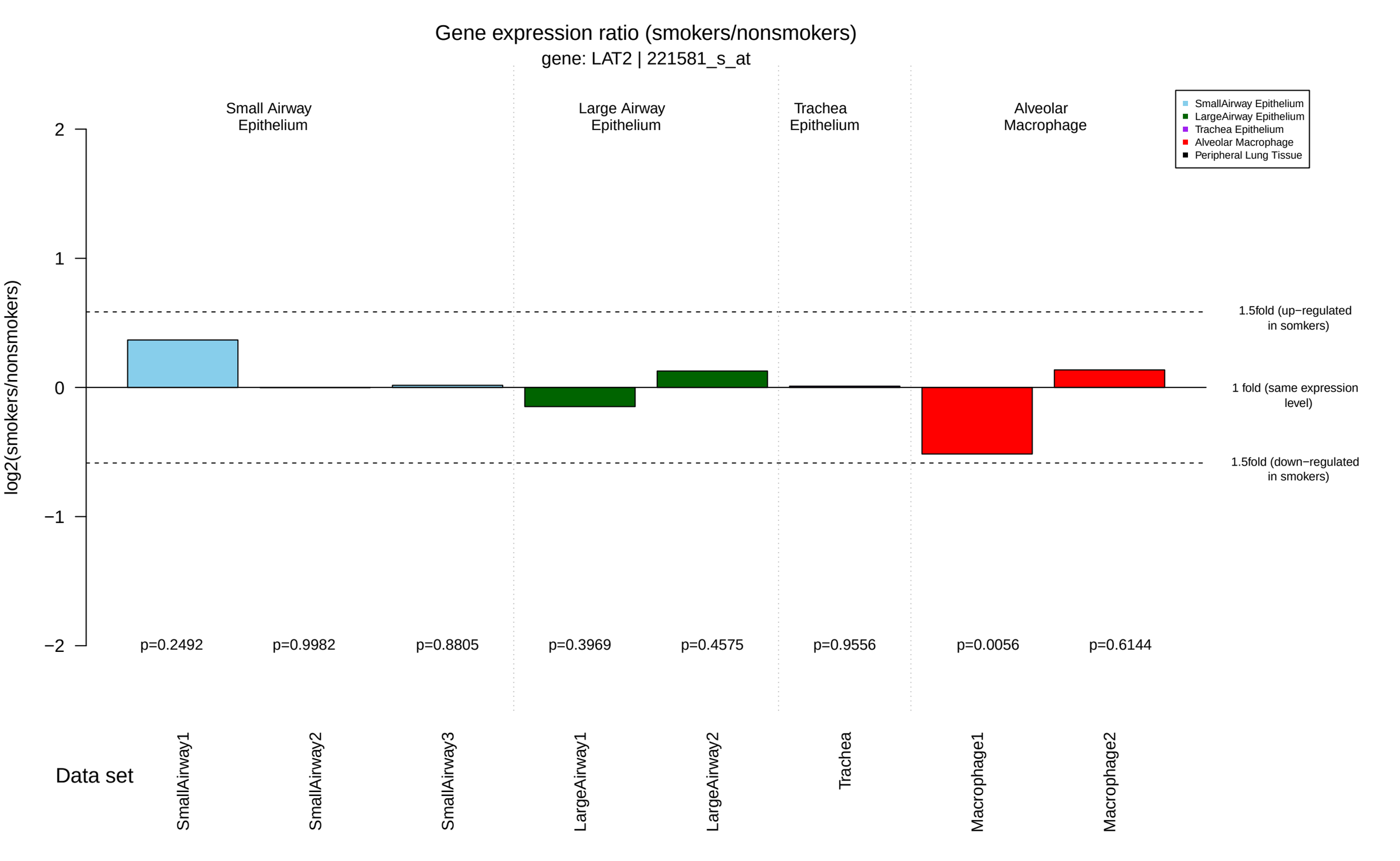
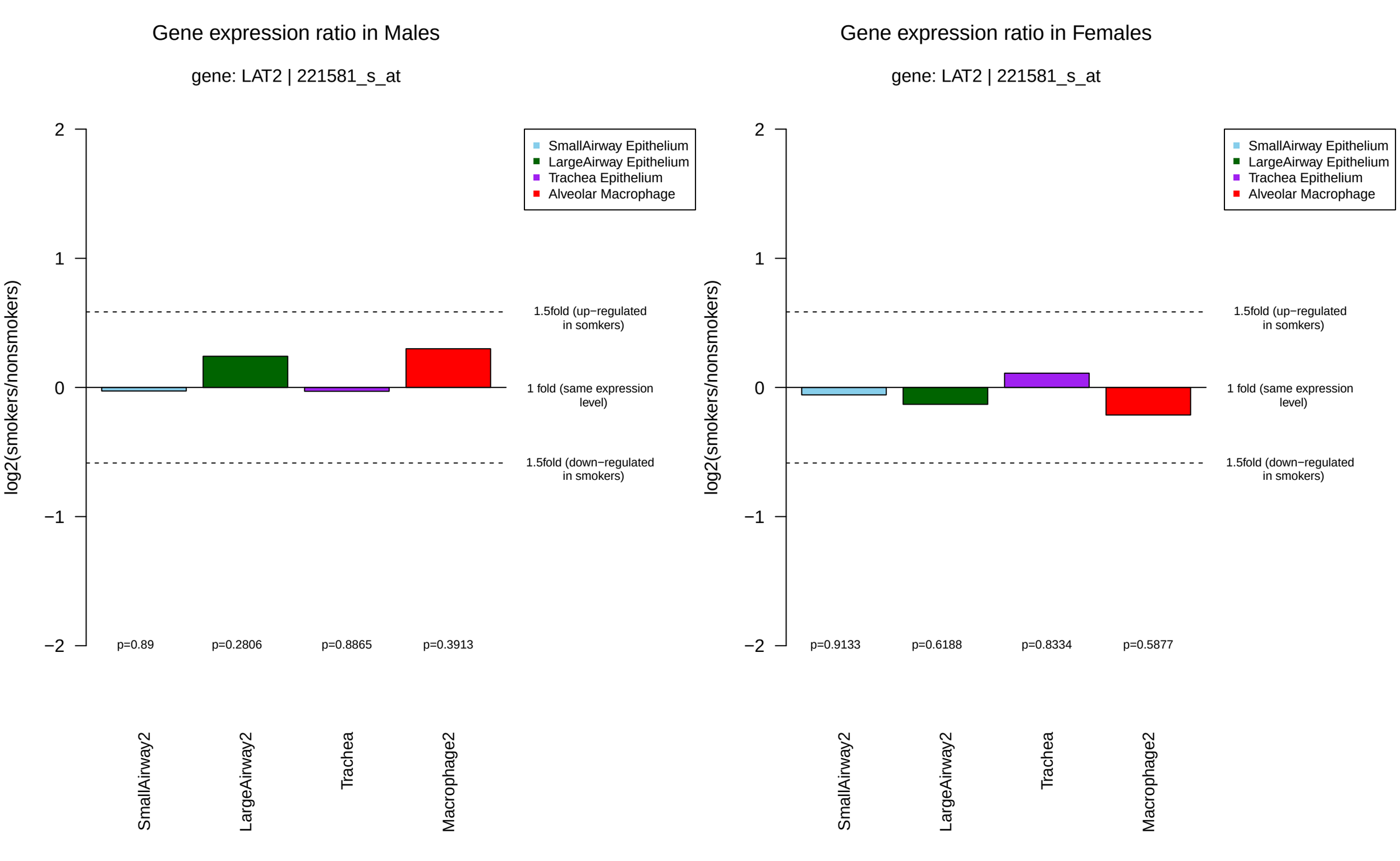
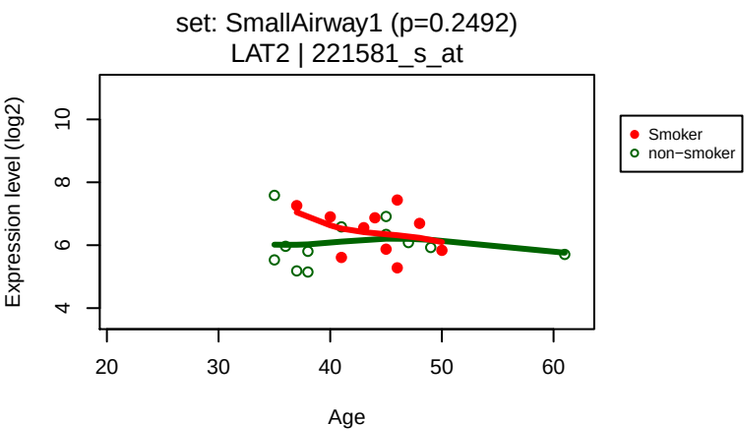
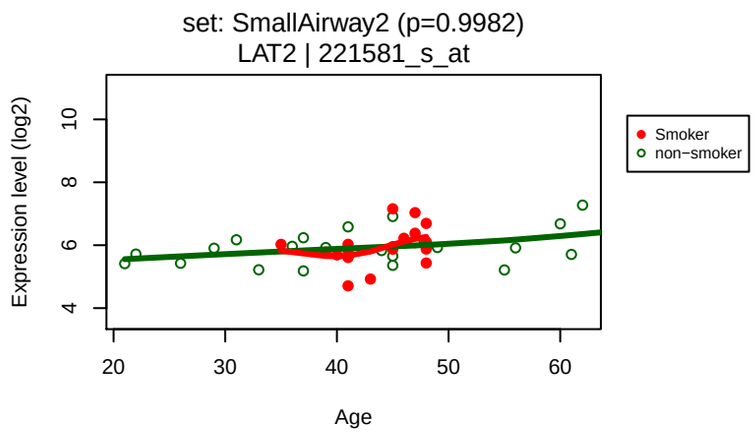
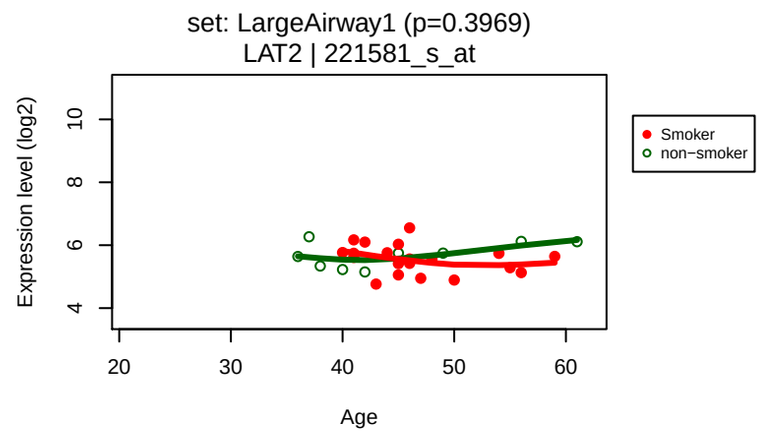

Gene expression of smokers and nonsmokers
gene: LAT2 | 221581_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2492
p=0.9982
p=0.8805
p=0.3969
p=0.4575
p=0.9556
p=0.0056
p=0.6144
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

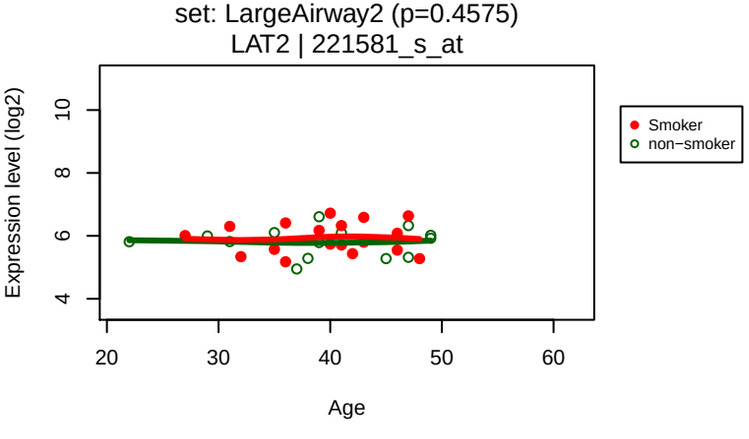

Gene expression ratio (smokers/nonsmokers)
gene: LAT2 | 221581_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2492
p=0.9982
p=0.8805
p=0.3969
p=0.4575
p=0.9556
p=0.0056
p=0.6144
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

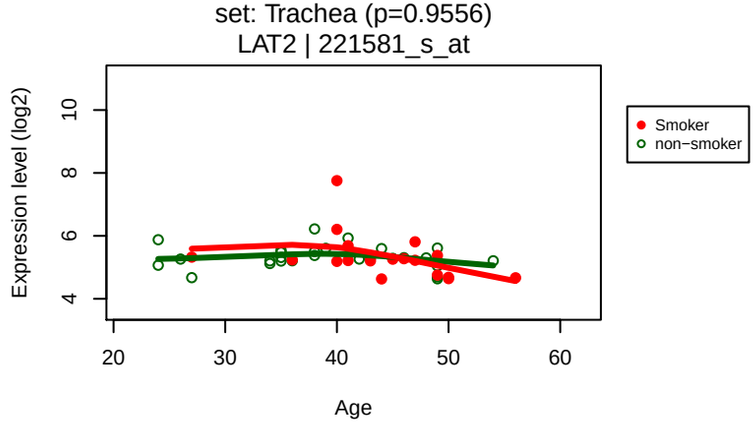

Gene expression ratio in Males
gene: LAT2 | 221581_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.89
p=0.2806
p=0.8865
p=0.3913
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LAT2 | 221581_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9133
p=0.6188
p=0.8334
p=0.5877
SmallAirway2
LargeAirway2
Trachea
Macrophage2

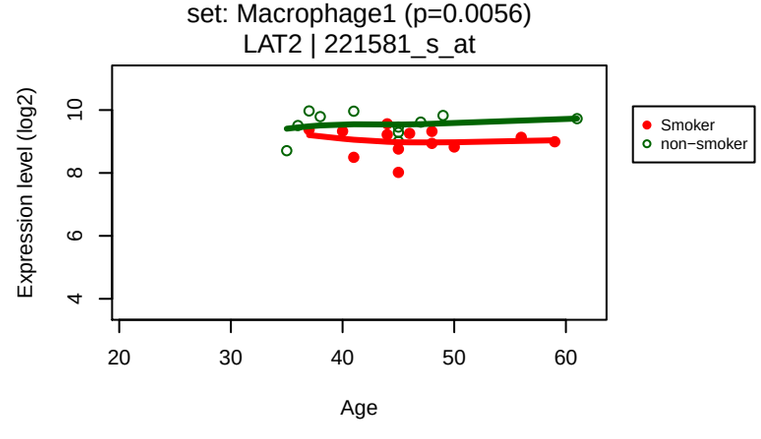

set: SmallAirway1 (p=0.2492)
LAT2 | 221581_s_at
Expression level (log2)
Age
Smoker
non−smoker

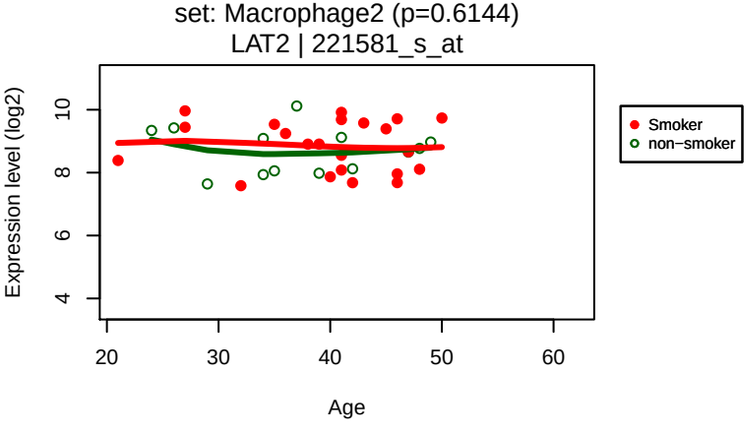

set: SmallAirway2 (p=0.9982)
LAT2 | 221581_s_at
Expression level (log2)
Age
Smoker
non−smoker

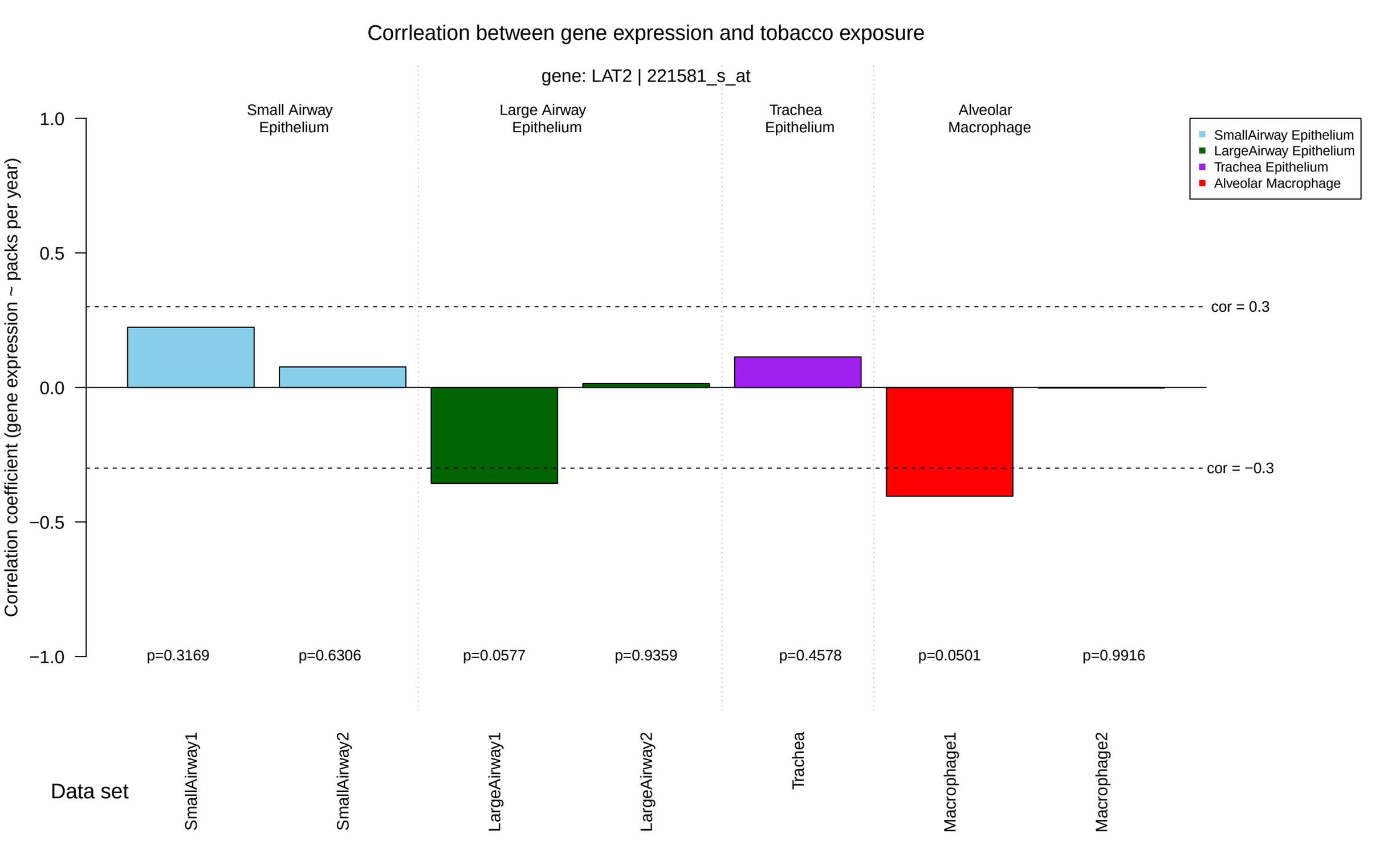

set: LargeAirway1 (p=0.3969)
LAT2 | 221581_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4575)
LAT2 | 221581_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9556)
LAT2 | 221581_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0056)
LAT2 | 221581_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6144)
LAT2 | 221581_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LAT2 | 221581_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3169
p=0.6306
p=0.0577
p=0.9359
p=0.4578
p=0.0501
p=0.9916
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2